NATIONAL CHILDREN'S SCIENCE CONGRESS (NCSC) 2026–27

Focal Theme: Science and Innovation for Sustainability
Sub-Theme: Food, Agriculture and Health

KRISHI-DRISHTI (■■■■-■■■■■■)

An Offline-First Edge-AI Plant Health Screening and Microclimate Monitoring System for Sustainable Agriculture

Category:	Senior Group (Class 9 / Age: 14–15 Years)
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"A practical school science project developed to provide low-cost, offline preliminary plant health screening for smallholder farmers in rural Bihar."

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5. PROJECT ABSTRACT

KRISHI-DRISHTI (■■■■-■■■■■■) is an ultra-low-cost (< ■1,500), 100% offline-first agritech system developed to provide preliminary foliar health screening and microclimate risk monitoring for smallholder potato and tomato farmers in rural Bihar. Over 35% of Solanaceae crop yields in North India are severely damaged annually by Late Blight (*Phytophthora infestans*) and Early Blight (*Alternaria solani*). Small marginal farmers (< 2 hectares) lack laboratory access and cannot rely on commercial cloud AI applications due to zero rural cellular data connectivity.

To address this, I designed a multi-tier physical architecture comprising: (1) a handheld controller wand (ESP32 DevKit V1) with active-low SCAN and PAGE buttons, an SSD1306 OLED screen, and a DHT11 canopy sensor; (2) a camera unit (AI-Thinker ESP32-CAM, OV5640 5MP sensor) fitted with a 3D-printed 8–10 cm cylindrical optical shroud to stabilize macro working distance and diffuse harsh sunlight; and (3) a headless offline edge neural co-processor running on a budget smartphone (Galaxy A03) over a standalone local Wi-Fi SoftAP mesh (192.168.4.1) requiring zero internet or SIM card.

The neural backbone uses MobileNetV3-Small quantized via Float16 post-training optimization to 2.03 MB (SHA-256: be71ededc...), achieving a mean inference latency of 4.94 ms on reference interpreter (~12–15 ms on budget mobile CPUs). Rigorous cryptographic auditing reconciled 7,710 dual-domain records into 6,965 unique training hashes, completely isolating holdout benchmarks. Across an independent 60-specimen botanical benchmark, the system achieved 88.33% strict six-class accuracy (53/60) and 92.98% accepted diagnostic accuracy (53/57, 3 gated UNCERTAIN), alongside 90.0% pathogen-family correlation (54/60). On 93 genuinely unseen outdoor PlantDoc images, raw baseline accuracy was 33.33% (31/93), triaged to 55.36% accepted diagnostic accuracy (31/56) by dual-stage Shannon entropy gating (H > 1.45 nats). Physical testing revealed real-world limitations: harsh torch drop-shadows induced false alarms on healthy foliage, and an alien tea leaf reached 87.0% false confidence inside the shroud, demonstrating that the shroud enforces focal geometry rather than out-of-distribution biological immunity. KRISHI-DRISHTI provides an honest, accessible preliminary screening tool that aids smallholder farmers in early cultural and organic intervention.

Keywords: *Edge-AI, Offline Screening, Solanaceae Blight, ESP32, MobileNetV3, Shannon Entropy, Microclimate Risk Advisory.*

6. INTRODUCTION

Agriculture is the primary livelihood for more than 70% of Bihar's population. Potato (*Solanum tuberosum*) and tomato (*Solanum lycopersicum*) are staple cash crops grown extensively by marginal farmers. However, every winter during the rabi season, foliar blights attack these crops. Under cold, foggy, and humid conditions, entire fields can be destroyed within 48 to 72 hours. When a small farmer loses their crop, their family loses their primary seasonal income.

7. WHY I SELECTED THIS PROBLEM

I chose this problem after visiting family and neighboring vegetable plots in rural Bihar. I watched local farmers spray expensive chemical fungicides indiscriminately without knowing whether the foliar spotting was Early Blight, Late Blight, or harmless physical scorching. One farmer spent over ■2,500 on an unverified fungicide cocktail that failed to save his potato crop. As a Class 9 science student, I wanted to build an affordable, portable electronic tool that a farmer could carry into the field for immediate preliminary screening without needing internet connectivity or laboratory fees.

8. LOCAL AGRICULTURAL RELEVANCE

Bihar's fertile alluvial plains (along the Ganga and Gandak belts) feature intensive potato and tomato farming. Most landholdings are under 2 hectares. These farmers face two major barriers: (1) **Zero Rural Cellular Connectivity:** In remote field tracts, 4G/5G signals are weak or absent, causing cloud-based apps to fail; and (2) **Economic Hardship:** Commercial diagnostic lab tests (■15,000–■30,000 per batch) are completely unaffordable for smallholders.

9. PROBLEM IDENTIFICATION IN THE FIELD

Through farmer surveys, I identified four key bottlenecks:
1. *Visual Confusion:* Early Blight (target-board spots caused by the fungus *Alternaria solani*) and Late Blight (water-soaked lesions caused by the oomycete *Phytophthora infestans*) look similar to farmers, leading to incorrect fungicide selection.
2. *Over-Spraying:* Panic spraying poisons soil biology, kills pollinators, and wastes household savings.
3. *Camera Inconsistency:* Uncalibrated smartphone photography suffers from macro blur, glare, and harsh shadows.
4. *Cloud Dependency:* Existing agritech systems assume reliable high-speed internet, excluding rural smallholders.

10. RESEARCH QUESTION

"Can a low-cost, portable handheld microcontroller wand integrated with local microclimate telemetry and an offline edge-AI engine provide reliable, preliminary foliar health screening for smallholder farmers without requiring internet connectivity or cloud servers?"

11. AIM AND OBJECTIVES

Aim: To design, build, program, and evaluate an ultra-low-cost (< ■1,500), 100% offline-first handheld diagnostic wand and microclimate risk monitoring system for early Solanaceae blight screening.
Objectives:
1. Build a dual-microcontroller interface (ESP32 DevKit V1 + ESP32-CAM) with non-blocking OLED, local SD storage, and DHT11 sensing.
2. 3D-print an 8–10 cm cylindrical optical shroud to standardize working distance and diffuse ambient shadows.
3. Quantize MobileNetV3-Small to Float16 precision (2.03 MB) for sub-15 ms inference on budget smartphones.
4. Implement Shannon entropy gating (H > 1.45 nats) to reject non-leaf objects, soil, and ambiguous frames.
5. Curate training data cryptographically with zero benchmark leakage, separating strict 6-class accuracy from pathogen-family correlation.

12. SCIENTIFIC HYPOTHESIS

Null Hypothesis (H■): Optical shroud distance constraint and canopy microclimate telemetry do not significantly improve offline neural classification or suppress false alarms compared to uncalibrated smartphone captures.
Alternative Hypothesis (H■): Standardizing optical acquisition geometry with a shroud, combined with Shannon entropy gating and microclimate risk advisory rules, will reliably triage foliar images, rejecting ambiguous non-leaf captures and enabling accurate preliminary screening on accepted frames under offline edge conditions.

13. BACKGROUND SCIENTIFIC PRINCIPLES

A. The Plant Disease Triangle: Infectious disease requires three concurrent factors: Susceptible Host + Virulent Pathogen + Favorable Environment. Even when *Phytophthora infestans* spores are present, they cannot germinate without canopy Relative Humidity (RH) > 85% at 18–24.5°C for several continuous hours (Wallin & Hyre models). Combining foliar visual screening with microclimate sensing is therefore far more robust than pixel classification alone.
B. Oomycete vs. True Fungus Biology: Late Blight is incited by *Phytophthora infestans* (an Oomycete water mold with cellulose/beta-glucan cell walls), while Early Blight is incited by *Alternaria solani* (a true Ascomycete fungus with chitinous cell walls). Chitin-inhibiting fungicides are completely ineffective against Late Blight. Distinguishing these pathologies prevents the purchase of wrong, costly chemicals.

14. EXISTING APPROACHES AND THEIR LIMITATIONS

Approach	Cost	Offline Capability	Failure Mode / Limitation
Visual Eye Inspection	Free	100% Offline	Subjective; cannot distinguish Early vs Late Blight in early stages.
Commercial Cloud Apps	Subscription / Ads	0% Offline (Needs 4G/5G)	Fails in rural fields with weak connectivity; uncalibrated camera blur.
Laboratory PCR / ELISA	■15,000 – ■30,000	N/A (Lab)	Takes 7–14 days; inaccessible to marginal farmers during outbreaks.
KRISHI-DRISHTI (This Work)	< ■1,500 Total BOM	100% Offline (Local Mesh)	Calibrated shroud, entropy gating, & microclimate advisory.

15. PROPOSED SOLUTION

KRISHI-DRISHTI combines three modular units: (1) an ergonomic handheld controller wand (ESP32 DevKit V1, SSD1306 OLED, DHT11); (2) an optical camera unit (ESP32-CAM, OV5640 5MP sensor, 8–10 cm 3D shroud, diffuse LED); and (3) a pocket edge-AI engine running on the farmer's smartphone via a local Wi-Fi SoftAP mesh (192.168.4.1), achieving 4.9 ms inference with zero internet.

16. COMPLETE SYSTEM ARCHITECTURE

[Farmer Presses SCAN] ■■> [ESP32 Controller Wand (GPIO 27)] ■■(Wi-Fi HTTP GET /api/capture)■■>
■■> [ESP32-CAM Unit] ■■(Saves UXGA JPEG to SDMMC) ■■(Broadcasts scan_id)
■■> [Pocket Android Smartphone (Local AI Daemon)] ■■(Pulls Frame & Runs MobileNetV3 TFLite + Shannon Entropy Gating)
■■> [Sends Diagnosis & Agronomic Prescriptions via /api/diagnosis/set]
■■> [ESP32 Controller Wand Polls /api/diagnosis/latest] ■■> [Renders 4-Page Carousel on OLED Screen]

17. HARDWARE COMPONENTS AND THEIR FUNCTIONS

Component	Specification	Interface / Pinout	Exact Functional Role
ESP32 DevKit V1	Dual-Core 240MHz, 520KB SRAM	Controller Core	Handles user buttons, FreeRTOS tasks, I2C OLED display, and Wi-Fi station telemetry.
AI-Thinker ESP32-CAM	ESP32-S + 4MB external PSRAM	SoftAP @ 192.168.4.1	Creates offline Wi-Fi SoftAP mesh; hosts web daemon; coordinates camera capture.
OV5640 Camera Sensor	5MP CMOS sensor (UXGA 1600×1200)	DVP 8-bit parallel	Captures high-resolution foliar images; configured for optimal dynamic range.
SSD1306 OLED Display	0.96 inch, 128×64 Monochrome OLED	I2C (SDA: 21, SCL: 22)	Displays 4-page diagnostic carousel: diagnosis, microclimate, organic cure, chemical cure.
DHT11 Sensor	0–50°C (±2°C), 20–90% RH (±5%)	Single-Wire (GPIO 18)	Monitors canopy microclimate; triggers environmental risk advisory interlock.
Push Buttons (SCAN / NEXT)	Tactile 12mm buttons, active LOW	GPIO 27 (SCAN), GPIO 26 (NEXT)	Triggers synchronous scan and cycles OLED carousel pages with software debouncing.
MicroSD Card Shield	8GB Class 10 MicroSD Card	SD_MMC 1-Bit DMA Mode	Stores raw captured JPEGs, diagnosis logs, and training telemetry records.
3D Optical Shroud	8–10 cm cylindrical black PLA baffle	Mechanical Wand Flange	Fixes working distance to lens sweet spot; diffuses lighting; suppresses ambient shadows.

18. SOFTWARE AND AI ARCHITECTURE

Model Architecture: MobileNetV3-Small pre-trained on ImageNet. It utilizes depthwise separable convolutions, inverted residual linear bottlenecks, and Squeeze-and-Excitation (SE) attention blocks.
• **Float16 Quantization:** Weights were quantized to IEEE 754 Float16, reducing model size from 4.02 MB to exactly **2.03 MB (2,123,856 bytes)** (SHA-256: be71ededc29f19f69ab98b63bebf281c6a1d88f468b6dc1a2a2dac655f406aaa).
• **Dual-Stage Gating:** (1) Foliar Plausibility Ratio rejects non-leaf surfaces (< 12% chlorophyll/necrosis); (2) Peripheral Dark Void Gate detects black cloths/dark drop-offs (> 45% dark border pixels); and (3) Shannon Information Entropy $H = -\sum p_i \ln p_i$. Ambiguous spreads ($H > 1.45$ nats or $p_{\max} < 0.50$) trigger an explicit UNCERTAIN advisory.

19. WORKING PRINCIPLE

1. **Placement:** Farmer places the shroud flush against the leaf.
2. **Trigger:** Farmer presses SCAN. The wand reads DHT11 temperature and humidity.
3. **Capture:** Wand triggers /api/capture on ESP32-CAM. The camera flashes, saves a 1600×1200 image to SD card, and buffers it in PSRAM.
4. **Inference:** Smartphone daemon pulls the image, resizes to 128×128, and runs TFLite inference in 4.94 ms.
5. **Gating & Risk Check:** Daemon evaluates foliar ratio and Shannon entropy, cross-referencing humidity risk.
6. **Display:** Wand OLED displays diagnosis. Pressing NEXT cycles: (1) Diagnosis & Confidence; (2) Microclimate Risk; (3) Organic Treatment; (4) Targeted Chemical Spray.

20. METHODOLOGY (ENGINEERING DESIGN CYCLE)

I followed five structured phases: (1) Field consultation with Bihar vegetable growers; (2) Dual-ESP32 breadboard prototyping and 3D printing; (3) Cryptographic dataset deduplication and holdout isolation; (4) Two-stage transfer learning and Float16 quantization; and (5) Multi-tier benchmarking across botanical specimens, held-out PlantDoc images, night-flash captures, and open-web samples.

21. DATASET PREPARATION AND QUALITY CONTROL AUDIT

Public agricultural datasets often suffer from hidden leakage. In the Kaggle New Plant Diseases Dataset, augmentations occurred prior to splitting. To guarantee integrity, I conducted a complete SHA-256 cryptographic audit:
• **Hash Reconciliation:** The 7,710 training records reconcile into exactly **6,965 unique training hashes** (6,600 clean PlantVillage paths + 1,110 clean PlantDoc paths [370 unique images replicated 3x]). The difference between 6,600 PV paths and 6,595 unique hashes was proven to be exactly 5 duplicate pairs inside upstream Kaggle folders with different filenames. A cryptographic assertion verified all 5 pairs share identical labels—yielding **zero cross-class conflicts** and **zero cross-dataset contamination**.
• **Holdout Isolation:** All 60 botanical benchmark leaves and all 93 held-out PlantDoc images were strictly excluded at both the SHA-256 hash level and perceptual dHash level (Hamming distance > 2).

22. TRAINING AND TESTING METHODOLOGY

Training was executed in Python using TensorFlow 2.21 using two-stage transfer learning:
• **Stage 1 (Warmup):** 2 epochs with frozen backbone using Adam (lr = 10■³), training dense classification head with Dropout (0.25).
• **Stage 2 (Fine-Tuning):** 2 epochs unfreezing top 25 convolutional layers (lr = 10■■) to adapt high-level features directly to foliar lesions.
• **Augmentation & Convergence:** Real-time affine rotations (±12%), horizontal flips, zoom (±8%), and contrast adjustments (±8%) were applied. The model reached 91.80% training convergence accuracy, 88.74% validation accuracy, and a balanced macro F1 score of 0.8863 across 2,850 validation images.

23. EXPERIMENTAL OBSERVATIONS

Physical testing revealed three critical empirical insights:
1. *Working Distance:* OV5640 requires 8–10 cm for sharp focus. Distances < 5 cm cause macro blur; distances > 15 cm capture background clutter. The shroud mechanically enforces 8–10 cm.
2. *Drop-Shadow Artifacts:* Nocturnal torch tests (10 images) showed that direct unshielded flash creates central specular hotspots with steep radial shadows. Convolutional filters mistook shadow borders for necrotic lesions, misclassifying 4/6 healthy leaves! This proved why diffuse lighting is essential.
3. *Out-of-Distribution Vulnerability:* An alien green tea leaf was classified as Potato Late Blight with 87.0% confidence inside the shroud. The shroud enforces focal geometry, but cannot act as a biological filter for unseen plants.

24. RESULTS AND THEIR SCIENTIFIC INTERPRETATION

KRISHI-DRISHTI reports results across multiple independent evaluation tiers, distinguishing between **Strict 6-Class Exact Match** and **Pathogen Family Correlation**:
• **60-Specimen Botanical Benchmark:** Achieved **88.33% strict six-class accuracy (53/60)**. Three specimens were gated as UNCERTAIN (95.0% coverage), yielding **92.98% accepted diagnostic accuracy (53/57)**. Pathogen-family correlation was **90.0% (54/60)**, as 1 Tomato Late Blight leaf was predicted as Potato Late Blight due to shared *Phytophthora infestans* biology (reported honestly as pathogen correlation, not an exact match).
• **PlantDoc Held-Out Field Benchmark:** Across 93 unseen outdoor field images, raw baseline accuracy was **33.33% (31/93)**. Dual-stage Shannon entropy gating safely rejected 37 ambiguous frames (39.78% rejection), raising accepted diagnostic accuracy to **55.36% (31/56)**.
• **Field Night-Flash Audit:** 1/6 in-domain tomato healthy verified; 3/4 alien OOD leaves (Hibiscus) safely intercepted by dark-void gating (75.0% safety interception).

25. COMPARISON TABLES AND GRAPHS

Evaluation Tier / Benchmark	Sample Size	Raw Baseline Accuracy	Accepted Diagnostic Accuracy	Gated / Uncertainty Rate
Production Daemon Verification	8 Canonical Leaves	100.0% (8/8)	100.0% (8/8)	0% (0/8)
60-Specimen Botanical Benchmark	60 Specimens (10/class)	88.33% (53/60)	92.98% (53/57)	5.0% (3/60 gated)
PlantDoc Held-Out Outdoor Set	93 Unseen Field Photos	33.33% (31/93)	55.36% (31/56)	39.78% (37/93 gated)
PlantDoc Full Adaptation Corpus	487 Outdoor Photos	43.12% (210/487)	66.25% (210/317)	34.91% (170/487 gated)
Field Night-Flash & OOD Audit	10 Real Field Photos	16.7% In-Domain (1/6)	50.0% Accepted (1/2)	75.0% OOD Gated (3/4)

Hand-Drawable Graph Reference (Figure 1: Accuracy vs. Coverage Under Dual-Stage Gating):
Plot a grouped bar chart with two bars per dataset:
• *Botanical Benchmark (60 leaves):* Bar 1 (Raw Accuracy) = 88.33%; Bar 2 (Accepted Accuracy) = 92.98% (Coverage = 95.0%).
• *PlantDoc Held-Out (93 field leaves):* Bar 1 (Raw Baseline) = 33.33%; Bar 2 (Accepted Accuracy) = 55.36% (Coverage = 60.22%).
• *Interpretation:* Shannon entropy gating filters out ambiguous or blurry frames, improving precision on accepted frames by +22.03% in field conditions.

26. REAL-WORLD LIMITATIONS AND KNOWN FAILURE MODES

Scientific integrity requires acknowledging limitations:
1. *Radial Shadows:* Direct unshielded flash causes steep shadows that mimic necrotic blight lesions. Shroud contact is necessary.
2. *Closed-Set Classification:* Alien non-crop leaves (e.g., tea leaf at 87% confidence) can fool the model. Closed-set models cannot recognize unseen plant species.
3. *Single-Point Telemetry:* An instantaneous canopy RH < 85% does not prove disease absence. It is an agronomic risk advisory to suppress chemical spraying, not definitive proof.
4. *Web Sample Scope:* Exploratory testing on 15 Wikimedia photos helped uncover failure modes, but does not replace large-scale field validation.

27. SAFETY AND ENVIRONMENTAL CONSIDERATIONS

• **Electrical Safety:** Operates at safe low DC voltages (3.3V and 5.0V from a power bank or 18650 cell with BMS). Zero AC shock risk.
• **Non-Destructive Testing:** The optical shroud contacts foliage gently without detaching or injuring the plant.
• **Ecological Protection:** By guiding farmers toward targeted organic remedies (Trichoderma viride, neem oil) and discouraging panic spraying, the project protects groundwater, soil microbes, and beneficial pollinators.

28. SUSTAINABILITY AND EXPECTED USEFULNESS

Aligns directly with NCSC 2026–27 sub-theme *'Food, Agriculture and Health'*:
• **Economic Sustainability:** At < ■1,500 total BOM, the wand costs less than a single bottle of proprietary fungicide. Avoiding one unnecessary spray pays for the device.
• **Zero Recurring Cost:** Runs 100% offline via local Wi-Fi SoftAP mesh without cloud API calls, data recharges, or subscriptions.
• **Food Safety:** Indiscriminate spraying leaves chemical residues on commercial vegetables. Timely organic interventions safeguard rural food supplies.

29. CONCLUSION

KRISHI-DRISHTI demonstrates that advanced Edge-AI and IoT microclimate sensing can be packaged into an ultra-low-cost, fully offline handheld tool for smallholder farmers. Decoupling camera capture, environmental sensing, and offline smartphone inference achieved 4.94 ms inference with zero internet. Shannon entropy gating elevated accepted field accuracy from 33.33% to 55.36%, while 88.33% strict accuracy was achieved on botanical benchmarks. KRISHI-DRISHTI is an honest, accessible preliminary screening tool that empowers small farmers to detect early distress and practice sustainable agriculture.

30. FUTURE IMPROVEMENTS (NEXT MAJOR MILESTONE)

Our defining next milestone is: **Large-scale physical field validation using images captured directly by the KRISHI-DRISHTI ESP32-CAM wand across rural agricultural plots in Bihar**. Future engineering enhancements include: (1) open-set recognition (OSR) to reject alien foliage; (2) digital leaf wetness sensing for multi-day Wallin/Hyre epidemiology; and (3) localizing the OLED and voice interface into regional dialects (Bhojpuri, Maithili, Hindi).

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